

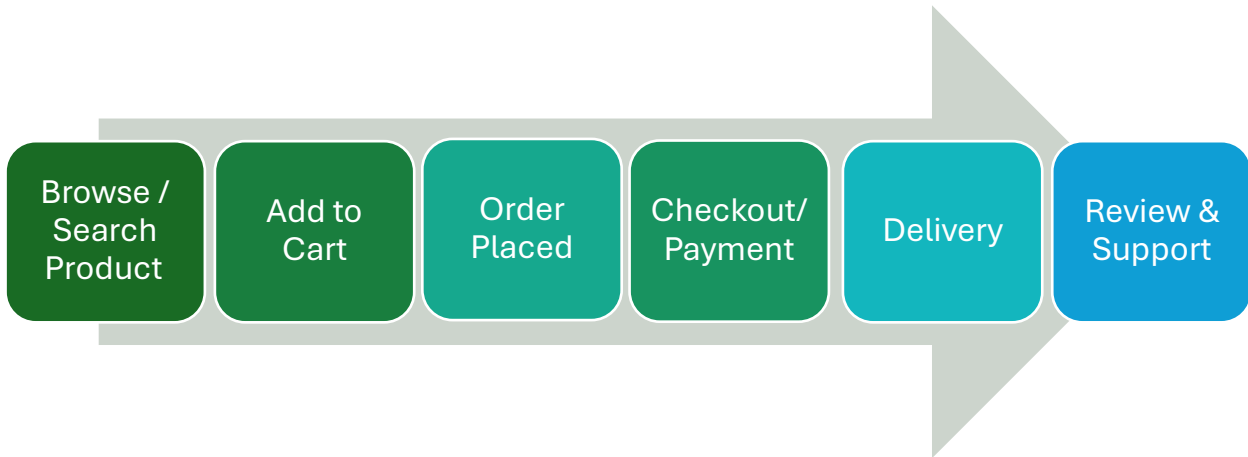
Week 1: Requirement Analysis and Customer Database Module

1. Requirement Analysis

Understanding E-Commerce Business

An e-commerce system manages the end-to-end flow of buying, selling, and managing products online. Below is a structured breakdown of the business process, customer data requirements, and entity definitions for system design and database modeling.

The Process Flow:



- **Browse / Search Product:** The Customer Searches for the product they want to purchase.
- **Add to Cart:** After choosing the product needed, they will add it to cart.
- **Order Placed:** The customer chooses the address to deliver and place the order.
- **Checkout/ Payment:** After Placing the order, the system make them to move to payment section and then checkout process.
- **Delivery:** After payment, the product will be assigned to the seller for dispatch and the product will be delivered.
- **Review & Support:** After successful delivery, the customer may give review about the product. For some product, they might need support for installation or return or replace.

Entities:

Entity	Description	Key Attributes	Key Relationships
Customer	End-user who browses and purchases products.	customer_id (PK), first_name, last_name, email, phone, created_at	One-to-Many with Order, Review
Seller	Merchant or vendor listing products on the platform.	seller_id (PK), company_name, email, phone, rating, bank_details	One-to-Many with Product

Entity	Description	Key Attributes	Key Relationships
Product	Item listed for sale on the platform.	product_id (PK), seller_id (FK), title, description, price, stock_quantity, sku, category	Many-to-One with Seller , One-to-Many with Review
Order	Transaction instance representing a customer purchase.	order_id (PK), customer_id (FK), order_date, status (pending, shipped, delivered, cancelled), total_amount, shipping_address	Many-to-One with Customer , One-to-One with Payment , Many-to-Many with Product (via <i>OrderItem</i>)
Payment	Monetary transaction tied to an order.	payment_id (PK), order_id (FK), payment_method (credit card, UPI, PayPal), status (success, failed, refunded), transaction_id, amount, payment_date	One-to-One with Order
Review	Feedback and rating submitted by a customer for a product.	review_id (PK), customer_id (FK), product_id (FK), rating (1–5), comment, created_at	Many-to-One with Customer , Many-to-One with Product

Customer Table Creation:

```
SQL> create table customer(customer_id varchar2(20) Primary key, first_name varchar2(20),last_name varchar2(20),email varchar2(30),phone number(10),created_at date);
Table created.

SQL> desc customer
Name                               Null?    Type
-----
CUSTOMER_ID                        NOT NULL VARCHAR2(20)
FIRST_NAME                         VARCHAR2(20)
LAST_NAME                          VARCHAR2(20)
EMAIL                              VARCHAR2(30)
PHONE                              NUMBER(10)
CREATED_AT                         DATE
```

Insert Values:

```
SQL> INSERT INTO customer
  2  VALUES ('C-1001', 'Ricky', 'Ponting', 'ricky200@edss.com', 9876543210, TO_DATE('2023-02-14', 'YYYY-MM-DD'));
1 row created.
```

```

SQL> INSERT INTO customer
2 VALUES ('C-100
3
SQL> INSERT INTO customer (customer_id, first_name, last_name, email, phone, created_at)
2 VALUES (
3 '&cust_id',
4 '&first_name',
5 '&last_name',
6 '&email',
7 &phone_num,
8 DATE '&created_date'
9 );
Enter value for cust_id: C-1002
old 3: '&cust_id',
new 3: 'C-1002',
Enter value for first_name: Calvin
old 4: '&first_name',
new 4: 'Calvin',
Enter value for last_name: Grey
old 5: '&last_name',
new 5: 'Grey',
Enter value for email: calvingrey@edss.com
old 6: '&email',
new 6: 'calvingrey@edss.com',
Enter value for phone_num: 8765432109
old 7: &phone_num,
new 7: 8765432109,
Enter value for created_date: 2024-09-23
old 8: DATE '&created_date'
new 8: DATE '2024-09-23'

1 row created.

```

Like this we can create as many rows we want.

```

SQL> select * from customer
2 ;

```

CUSTOMER_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE	CREATED_A
C-1001	Ricky	Ponting	ricky200@edss.com	9876543210	14-FEB-23
C-1002	Calvin	Grey	calvingrey@edss.com	8765432109	23-SEP-24
C-1003	sarath	Chandar	sarathchad@edss.com	9087654321	09-NOV-23
C-1004	Amar	Mahtani	amar23@edss.com	8901234567	09-SEP-25

Update:

```
SQL> update customer set phone = 9870654312 where customer_Id = 'C-1004';  
1 row updated.
```